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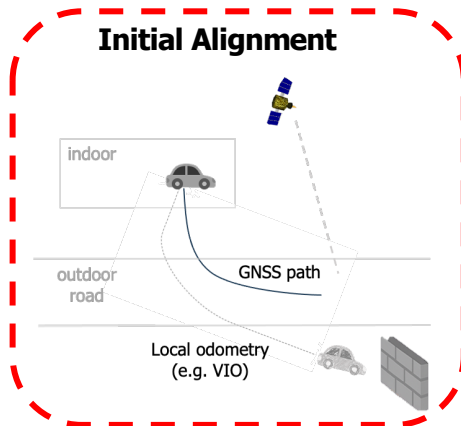


Certifiable Navigation

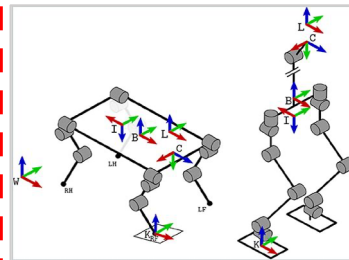
Initial Positioning



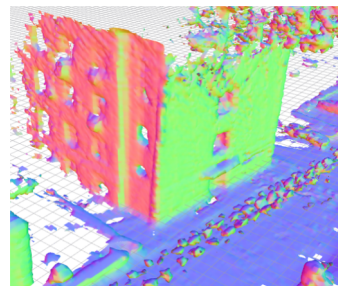
Initial Alignment



Sensor Calibration



Localization and Mapping



Certifiable GNSS/Local Frame Alignment using Doppler measurements

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Spe 29, 2025 @ Honor Workshop, Hong Kong



Toward Alpha plan: intelligent navigation

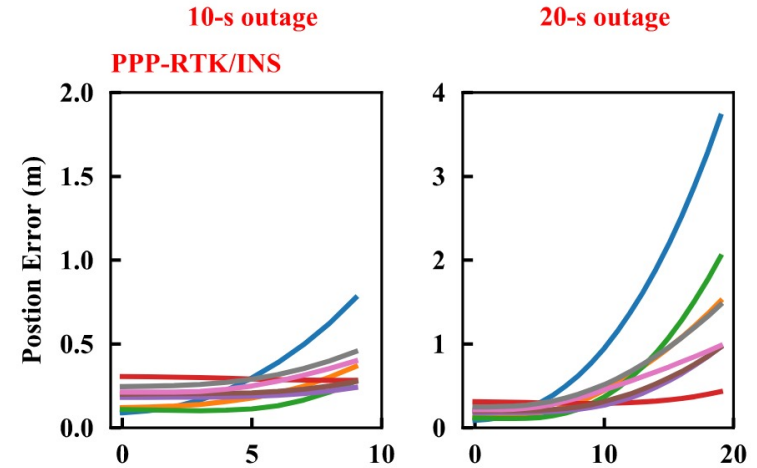


- ✓ **Various demand** in scenes and robots
- ✓ **Safe, precise and continuous** navigation is required
- ✓ **Seamless** navigation across all scenes is required

Seamless navigation challenge: tunnel



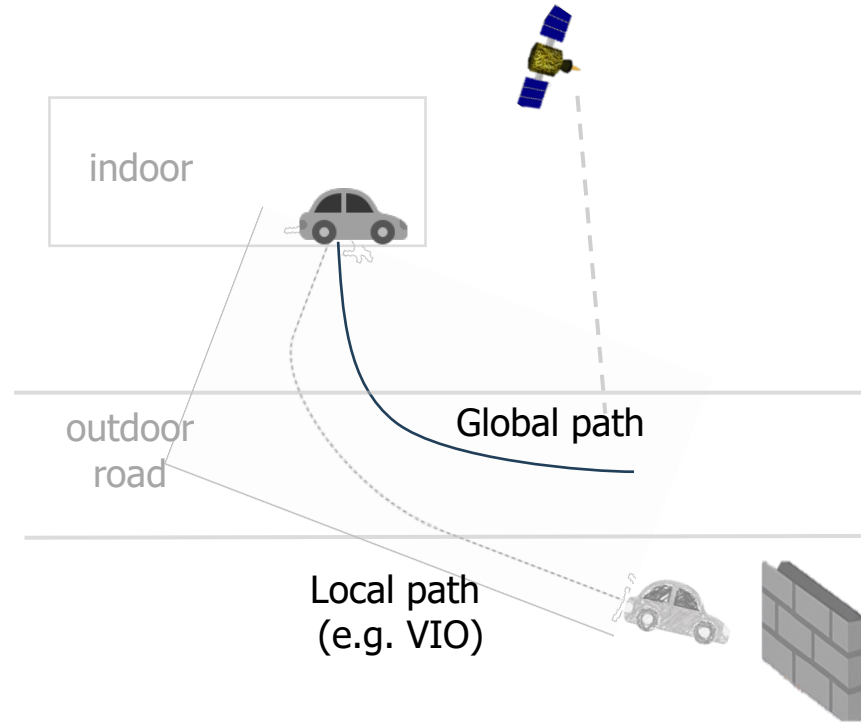
Driving through a tunnel



Position error with GNSS outage
[Li et.al., 2022 *T-TITS*]

- ✓ A frequent initial alignment for GNSS/local frame is necessary!

Seamless navigation challenge: urban parking



Dangerous!

Representative Work	Sensors	Input	Variable	Estimation	Dynamics	Initial guess
Gade(2016), Eling (2013)	GNSS	Multi-antenna	Yaw	Closed-form	Low	X
Wu (2011)	GNSS-IMU	Acceleration	Yaw	Closed-form	High	X

Gap 1: Rely on
Abundant satellites

Gap 2: Require
good initialization

Is it possible to obtain a guaranteed **optimal** estimate with **limited** satellite?

Doppler based alignment

problem ^[1]:

$$\min_{\mathbf{R}_w^e \in SO(3), \bar{t} \in \mathbb{R}} \sum_{i=1}^K \|z_i\|_\epsilon^2$$

$$z_i = (\mathbf{R}_w^e \mathbf{v}_{r,i}^w - \mathbf{v}_{s,i}^e)^T \mathbf{n}_i + \bar{t} - \bar{D}_i$$

ECEF-world
rotation

Local
receiver
velocity

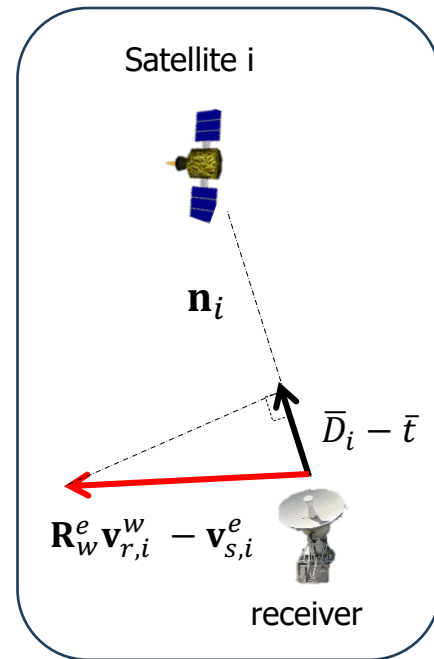
Satellite
velocity

Line of
sight
vector

Receiver
clock
shift rate

Doppler
measurement

- ✓ Current feasible region $SO(3)$ is **non-convex**
- ✓ There are multiple **local optima** in a non-convex problem

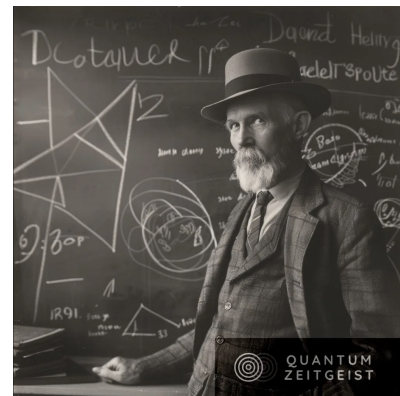


How to guarantee optimality?

Hilbert's 17th problem: Can we rewrite a non-negative polynomial as sum of squares (SOS)?

Answer: [J. Lasserre, 2001]

For some problems, we can try to **guarantee optimality** of $f(x)$ by testing the **positivity constraint** $h(x)$ using SOS.



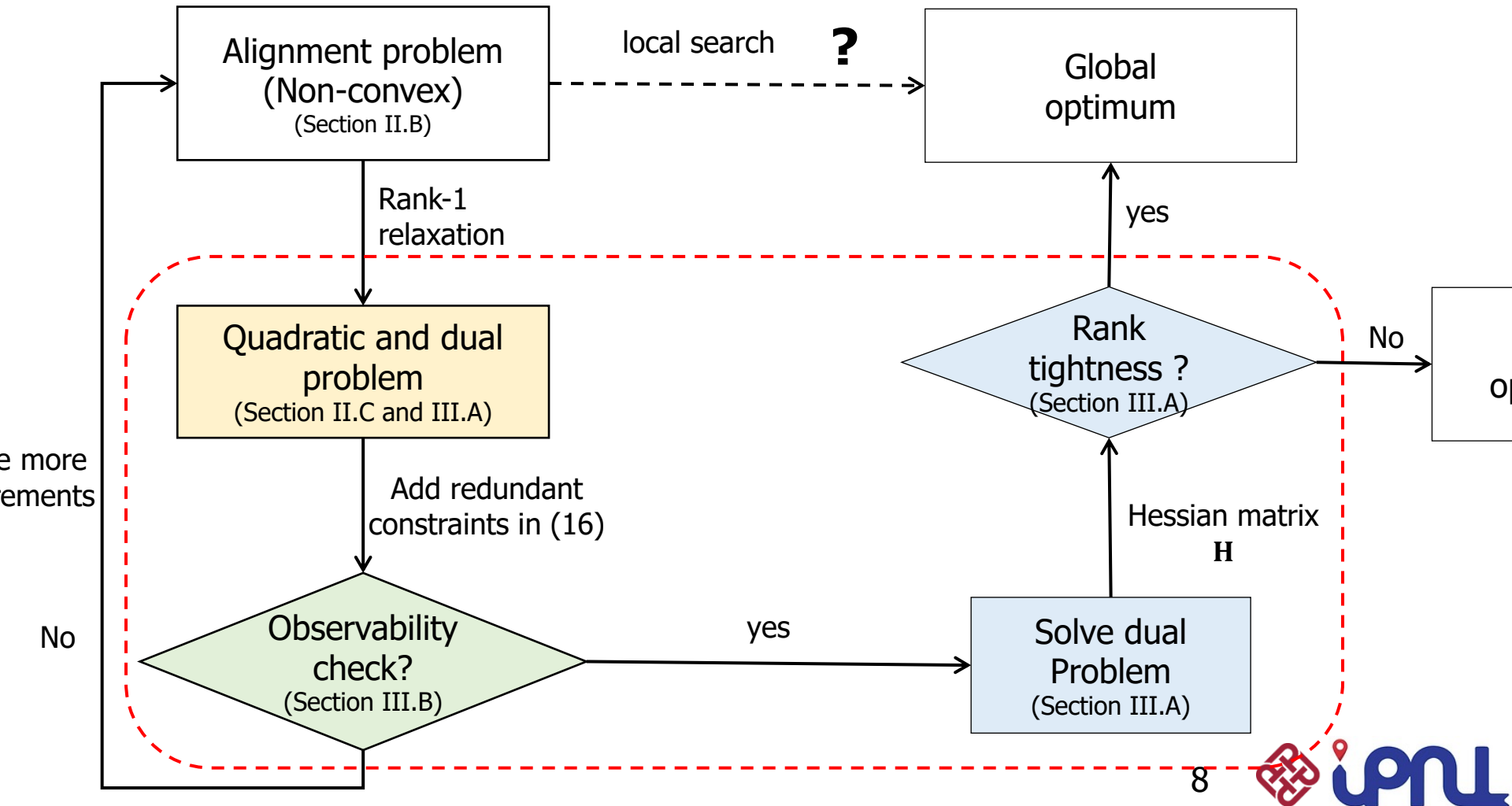
David Hilbert
(Ref: QUANTUM)

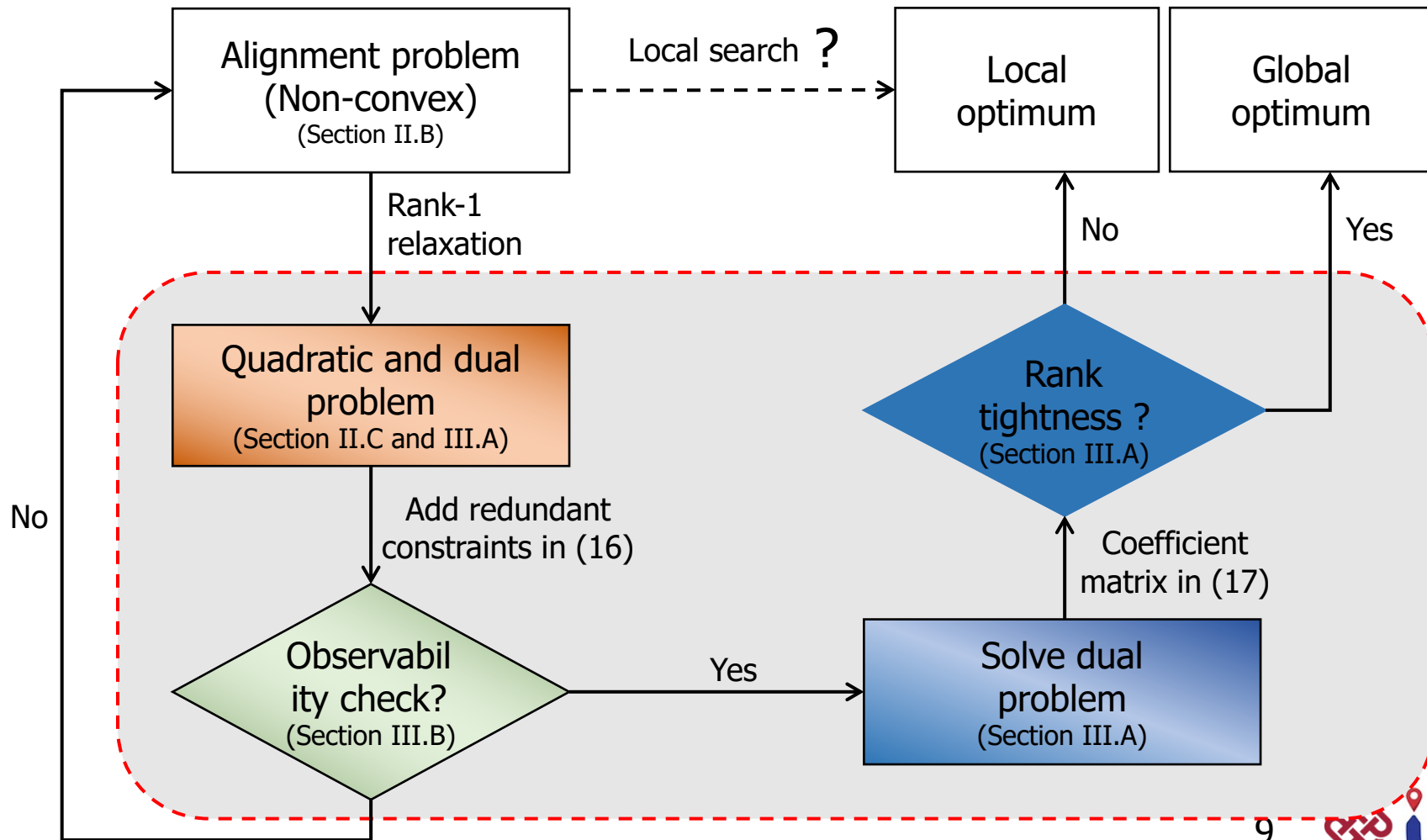
$$\min f(x)$$

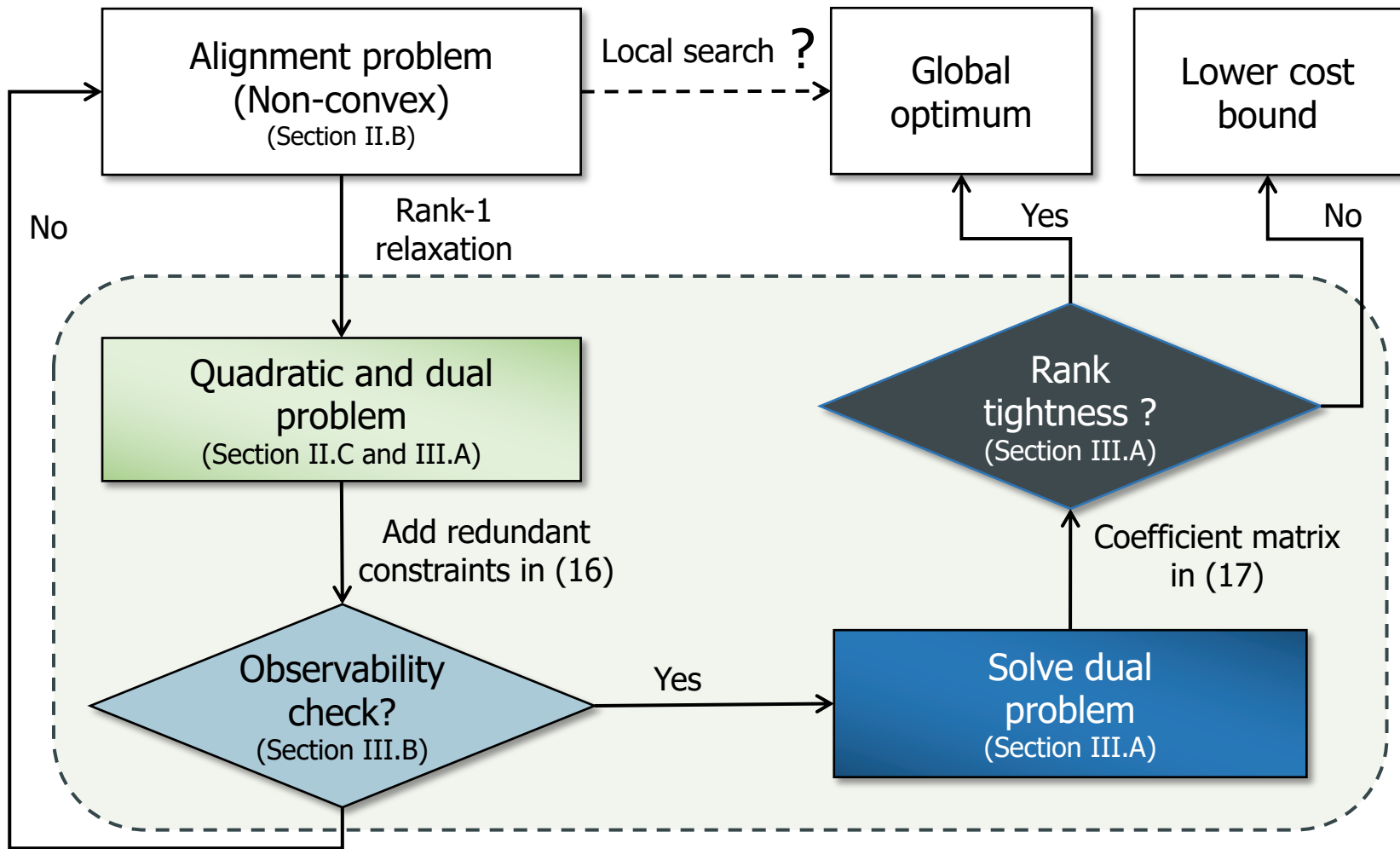


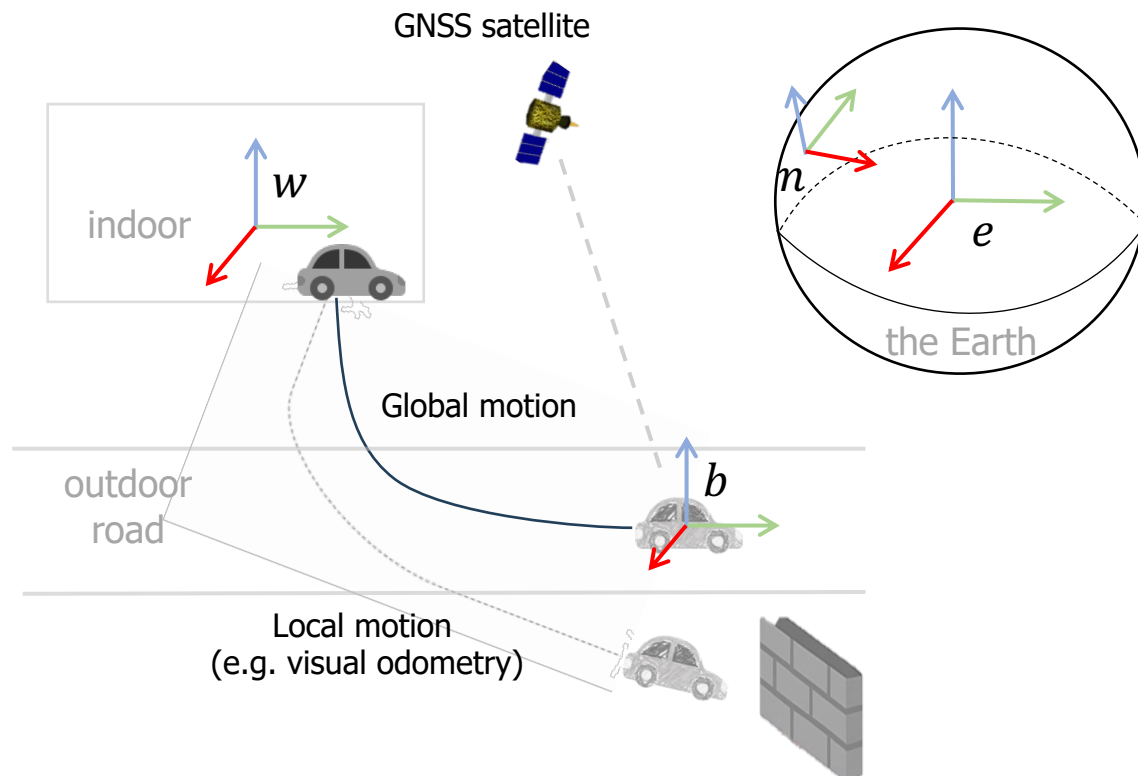
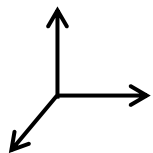
$$h(x) = f(x) - \gamma \geq 0 \quad \longrightarrow \quad h(x) = \sum_{i=1}^k g_i(x)^2 \quad \longrightarrow \quad h(x) \geq 0$$

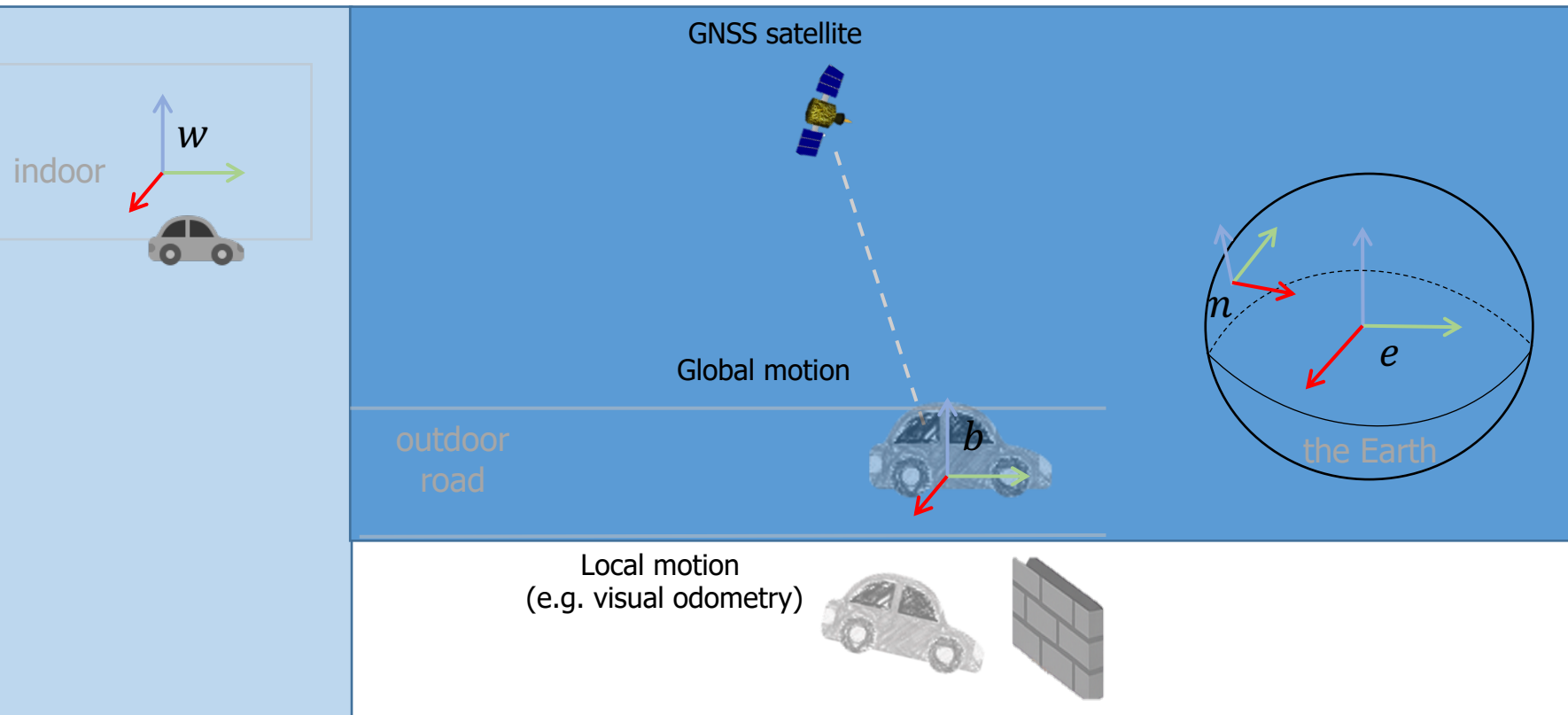
with given lower bound γ







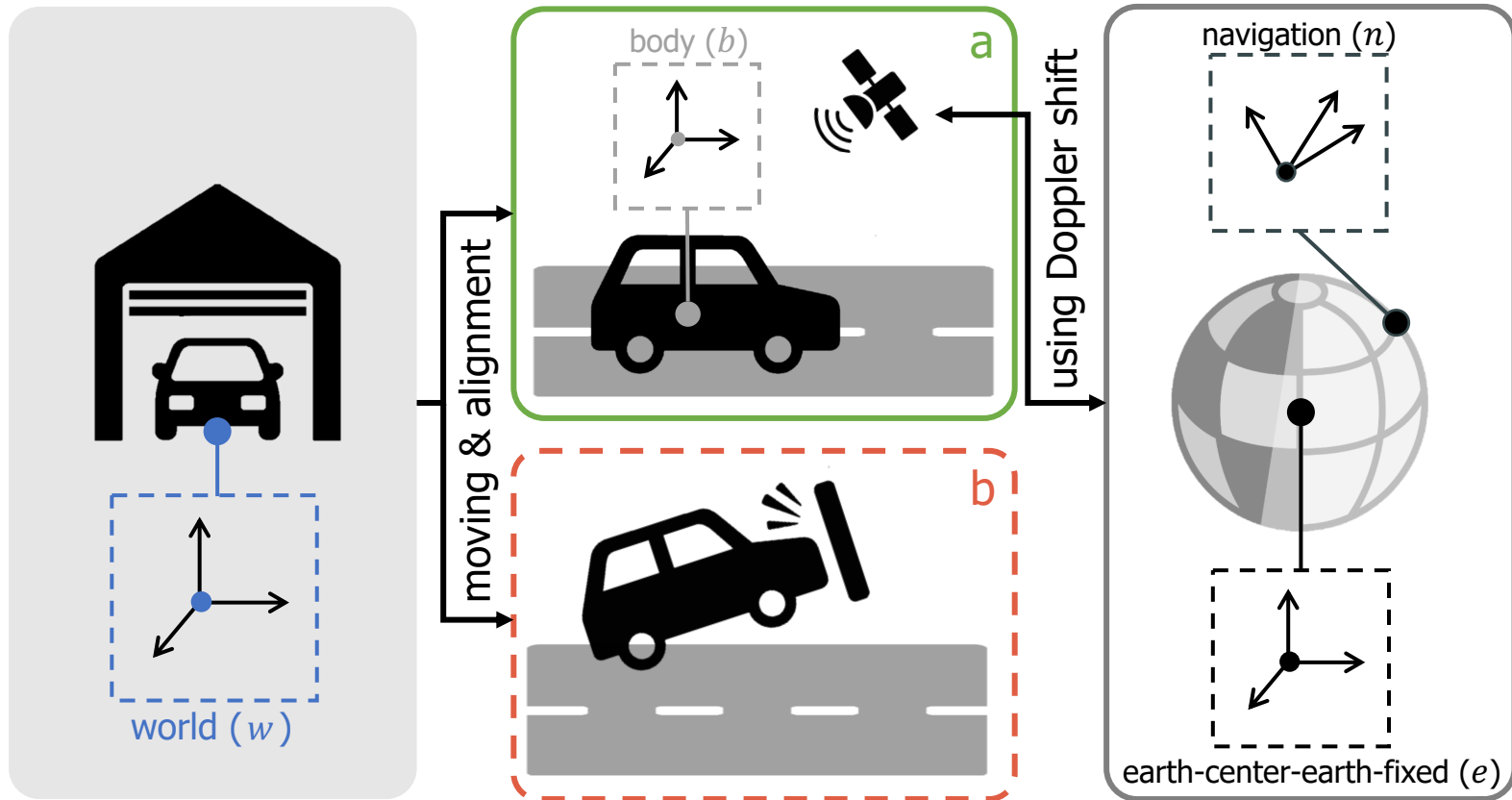




Indoor

Outdoor

the Earth





Step 1: problem reformulation

	Original (non-convex)	Quadratic (non-convex)	Dual (concave)
Variables	$\mathbf{R}_w^e \in SO(3), \bar{t} \in \mathbb{R}$	$\mathbf{r} = \text{vec}(\mathbf{R}_e^w) \quad \mathbf{x} = [\mathbf{r}^T \quad y]^T$	λ, γ, M, N
Cost function	$\min_{\mathbf{R}_w^e \in SO(3), \bar{t} \in \mathbb{R}} \sum_{i=1}^K \ z_i\ _c^2$	$\min_{\mathbf{x}} \mathbf{x}^T \bar{\mathbf{Q}} \mathbf{x}$	$\max_{\lambda, \gamma, M, N} \gamma$
Constraints	$z_i = (\mathbf{R}_w^e \mathbf{v}_{r,i}^w - \mathbf{v}_{s,i}^e)^T \mathbf{n}_i + \bar{t} - \bar{D}_i$ $SO(3) \equiv \{\mathbf{R} \in \mathbb{R}^{3 \times 3} :$ $\mathbf{R}^T \mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1\}$	$s.t. \mathbf{R}^T \mathbf{R} = y^2 \mathbf{I}_3,$ $\mathbf{R} \mathbf{R}^T = y^2 \mathbf{I}_3,$ $\mathbf{R}^{(i)} \times \mathbf{R}^{(j)} = y \mathbf{R}^{(k)},$ $i, j, k = \text{cyclic}(1, 2, 3), y^2 = 1$	$s.t. \bar{\mathbf{Q}} + \mathbf{P}(\lambda, M, N, \gamma) \succeq \mathbf{0}$

Key insight here is to **solve the Dual problem** and **recover original solution**.

Step 2: observability check

Lemma 1 (Necessary conditions of observability with abundant constraints). *The problem for the dual problem with abundant constraints is observable with the Hessian matrix's degree of freedom (DOF) ≥ 4 . To obtain instantaneous alignment, we need at least 2 satellites and velocity along 2 axis.*

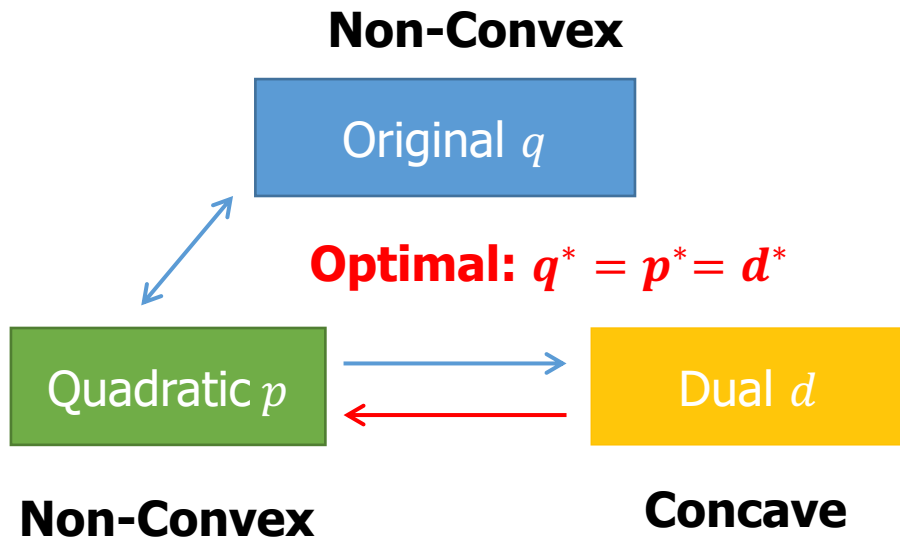
$$M_{1:K} = [\mathbf{v}_i \otimes \mathbf{n}_i]_{i=1}^K = \mathbf{V} \odot \mathbf{N}$$

Local receiver Velocity groups Hadamard product Line of sight vector groups

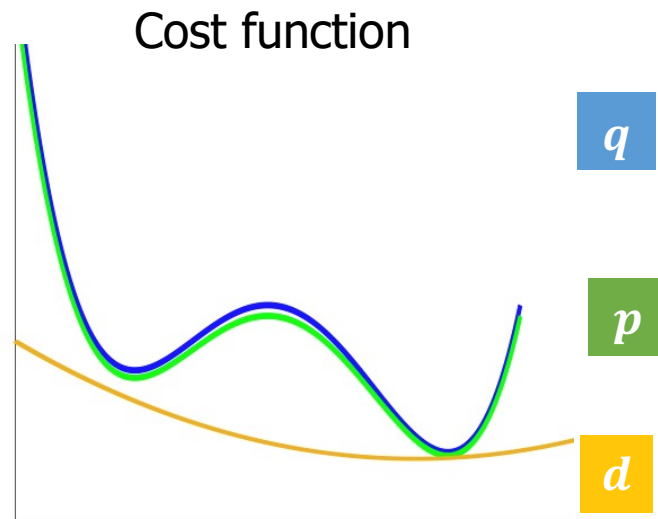
$4 \leq \text{rank}(\mathbf{M}) \leq \text{rank}(\mathbf{V}) \cdot \text{rank}(\mathbf{N})$

$\text{rank}(\mathbf{V}) \geq 2 \text{ and } \text{rank}(\mathbf{N}) \geq 2$

Step 3: Optimality guarantees



Strong duality:
Karush-Kuhn-Tucker (KKT) condition



Tight relaxation:
Rank-1 condition

Compared methods

Configurations	VOBA ^[1]	GVINS ^[2]	Our method (SDP)
Solver	SPV+SVD	weighted least squares	CVX ^[3]
Iteration threshold	No	10^{-6}	Best precision
Solution type	Closed-form	Heuristic	Post-hoc Guaranteed
Initialization	No	Yes	No

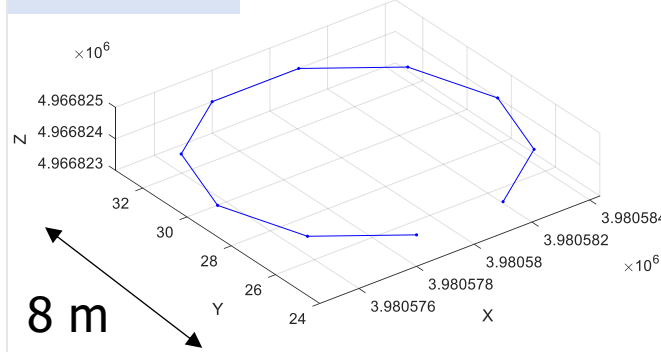
[1] Zhang et.al., *IEEE SJ*, 2020

[2] Cao et.al, *T-RO*, 2021

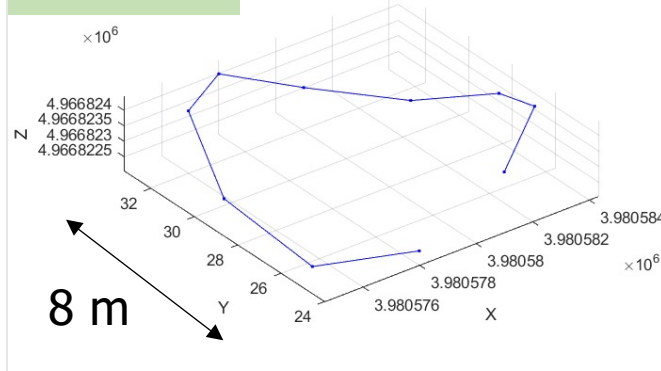
[3] Boyd, convex optimization, 2006

Simulation tests

2d motion

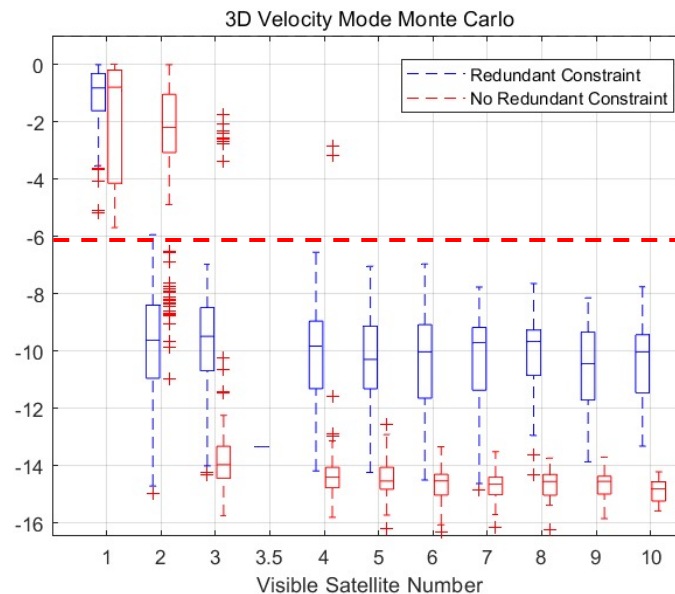
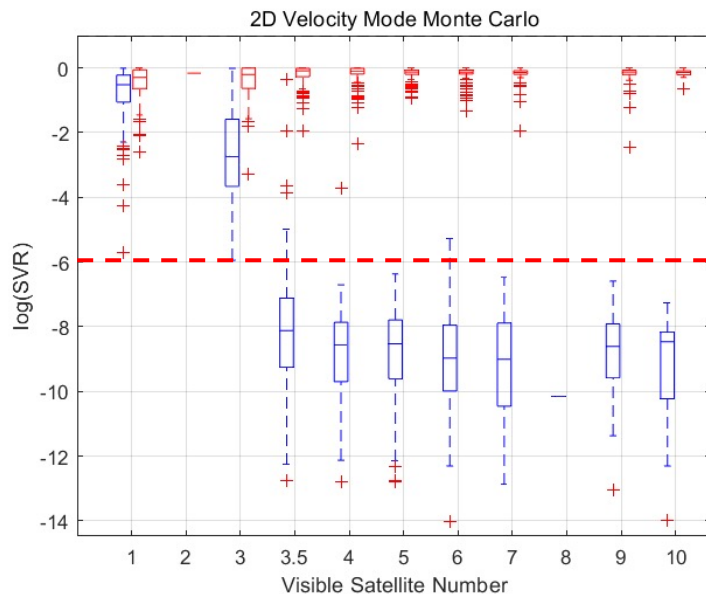


3d motion



Settings	Value
Interval	1 second
Duration	10 seconds
Signal Frequency	1575.42 MHz
Receiver velocity	3 m/s
Orbit type	Walker
Clinical angle	55°
Elevation angle	≥10°

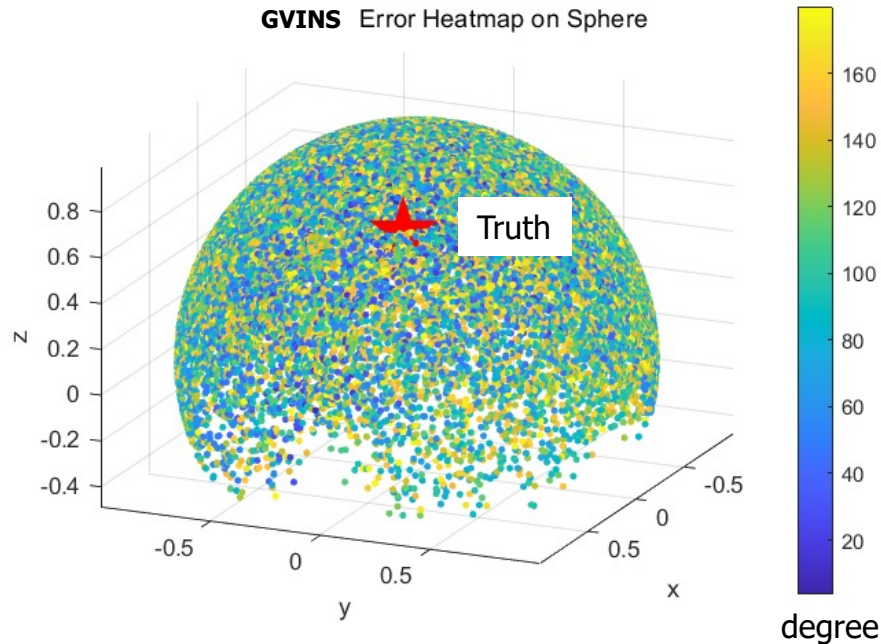
Constraints v.s. Rank-tightness



optimal

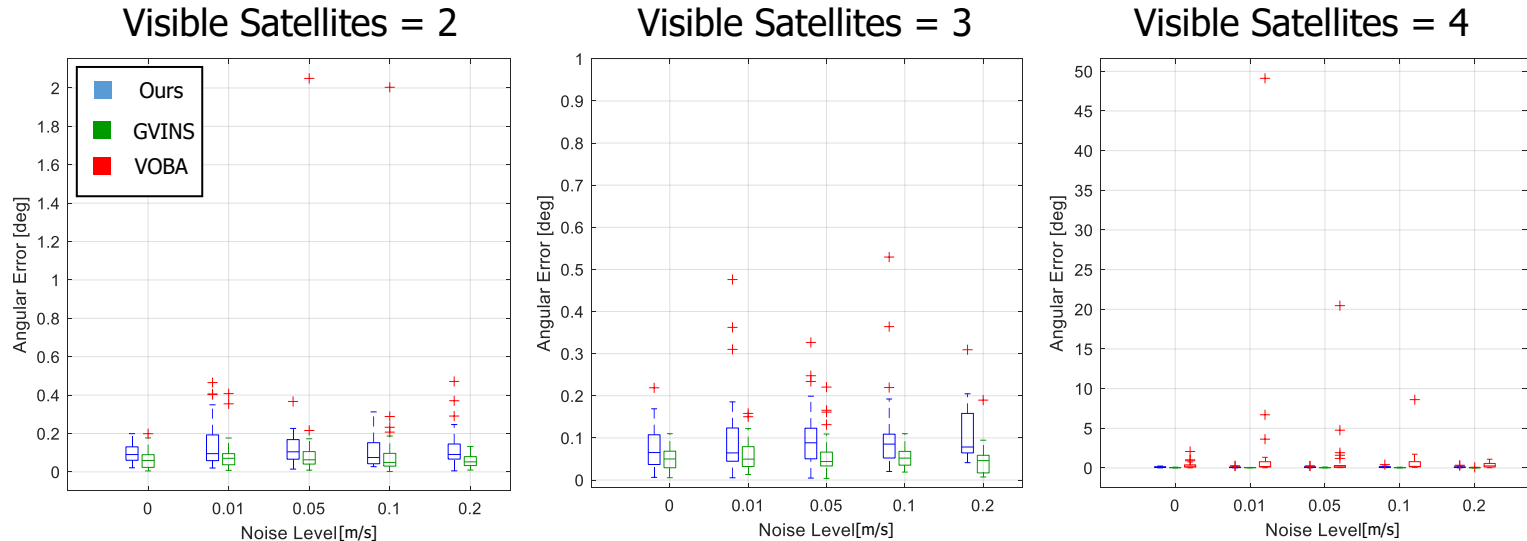
1. Redundant constraints increase the **robustness** of rank-tightness.
2. **Without abundant constraints, 2D motion** always leads to **failed** alignment.

Initial guess v.s. Local searching method



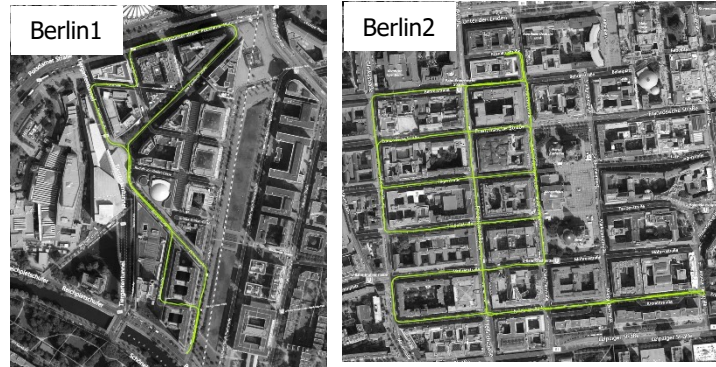
Under **poor observability**
(visible satellite number = 2),
Gauss-Newton is easy to be
stuck in **local optima** with
random initialization.

Doppler noise v.s. Alignment errors



1. SDP provides certifiably optimal solution but the **precision** of SDP is worse than GN, which is caused by the solver.
2. **VOBA** outputs closed-form solution but is fragile and only work when the **satellite number ≥ 4** .

Real world tests



	Hardware	Features
Ground-truth	NovAtel SPAN	Yaw only
GNSS receiver	U-blox M8T	L1
GNSS Antenna	Patch antenna	GPS, GLONASS
odometer	CAN	velocity and yaw-rate

Real world results

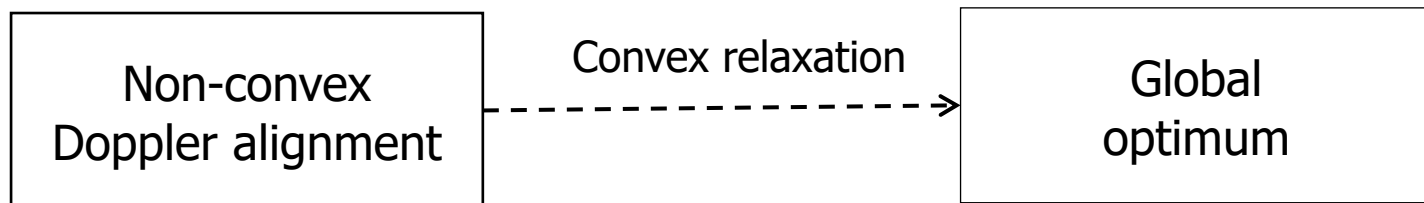
With 3 LOS satellites, motion in 60 s with two turning

Yaw error [°]	VOBA	GVINS	Ours
Berlin	18.06	12.08	12.07
Berlin-10	17.12	200.32	9.36

1. With down-sampled sparse measurements, **GVINS** is stuck in **local optima** given random initialization, while VOBA and Ours is not affected;
2. **Our method** achieve the **best precision** compared to other methods.

Take-aways

- Certifiably optimal GNSS/Local frame alignment
 - Certifiably correct alignment with at least **2 satellites** (under 3D motion + abundant constraints);
 - Both observability and **optimality** analysis is performed.
 - Codes and data will be **open-sourced**.
- Meaning of this work



Future work

- Limitation

1. Ignore **measurement outlier** for simplicity;
2. The SDP **optimization accuracy** is limited;

- Future work

1. Develop **robust and certifiable using M-estimation and Lagrangian duality**;
2. Develop a fast and accurate certifier **without SDP optimization**.

References

- [1] Li, X., Qin, Z., Shen, Z., Li, X., Zhou, Y., & Song, B. (2022). A high-precision vehicle navigation system based on tightly coupled PPP-RTK/INS/odometer integration. *IEEE Transactions on Intelligent Transportation Systems*, 24(2), 1855-1866.
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- [3] P. D. Groves, *Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems*. Norwood, MA, USA: Artech House, 2013.
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- [6] Q. Zhang, S. Li, Z. Xu, and X. Niu, "Velocity-based optimizationbased alignment (VBOBA) of low-end MEMS IMU/GNSS for low dynamic applications," *IEEE Sensors J.*, vol. 20, no. 10, pp. 5527–5539, May 2020.
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- [9] Li, T., Pei, L., Xiang, Y., Zuo, X., Yu, W., & Truong, T. K. (2023). P 3-LINS: Tightly coupled PPP-GNSS/INS/LiDAR navigation system with effective initialization. *IEEE Transactions on Instrumentation and Measurement*, 72, 1-13.
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- [11] J. Lasserre, "The Moment-SOS hierarchy," Aug. 10, 2018, arXiv: arXiv:1808.03446. doi: [10.48550/arXiv.1808.03446](https://doi.org/10.48550/arXiv.1808.03446).

Open Source & Contact Information

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 Open-sourced software:
LEO positioning; FGO-KFV comparison.



<https://github.com/Baoshan-Song/Certifiable-Doppler-alignment>

(mathematical derivation, codes, data)

